

The salty ocean breeze blew through the windows of the operating van and onto Private George Elliott's cheeks. Looking out from Opana Point, a location 532 feet above sea level, there was an unobstructed view of the Pacific waters. Even in December, it was another welcoming, cloudless day at the tropical home of the U.S. Pacific Fleet.

But right now, it is not the time for Elliott to get distracted. Joseph Lockard, the more experienced radar operator, was teaching Elliott how to use the radar oscilloscope. Two years ago, six mobile sets of this long-range aircraft search apparatus called the SCR-270 radar had been installed across Hawaii. Each set consisted of four trucks that carried a transmitter, modulator, water cooler, receiver, oscilloscope, operator, and generator. Along with that came a 45-foot tall, rotating, metal antenna that emits and receives radio waves. This device could detect planes and ships before the naked eye sees it. On the morning of December 7, 1941, the Opana Radar Station was the only radar station in operation. The higher-in-command officers limited the use of the radar sets, afraid that the equipment would wear out.

"It seems like the truck's running late," Lockard stifled a yawn as he observed Elliott work the oscilloscope. They had been here since 4:00 in the morning. Usually, a truck came around 7:00 to drive them to breakfast. Lockard glanced at the clock. It read 7:02.

Oh, well, Lockard thought. *A little extra practice never hurts anyone.*

Suddenly, out of the oscilloscope's vibrating signal line emerged something unusual. It was a blip. Startled, Elliott adjusted the focus and intensity knob controls to gain a clearer view of the signal.

“Check this out,” Elliott called to his companion. “The antenna’s receiving an enormous blip.”

Elliott got up and let Lockard settle in front of the oscilloscope screen. Lockard tweaked the range dial and the vernier, lining up the left edge of the echo signal with the hairline.

“It’s moving too fast to be a ship,” Lockard squinted and bit his lip. “It could be some planes in the air.”

The ruler below the screen, which was scaled in miles, told that the echo signal was 137 miles north of Oahu. It was closing in on the island at such a high speed that was unlike anything the operators had seen before. They agreed to double check the equipment for malfunctioning parts.

“The receiver and transmitter are working well,” Elliott confirmed. He was now certain the blip meant something.

“So are the antenna and generator!” Lockard hopped back into the operating van. He attached the long viewing tube onto the oscilloscope screen, leaned over, and peered into the narrow eyepiece. The blip was nearing the output signal fast, at about 180 miles per hour. Elliott plotted the coordinates as Lockard called them out. The two operators knew this needed to be reported. Immediately.

“I need Fort Shafter,” Elliott phoned the switchboard operator at the information center, who was Joseph McDonald at the time. McDonald informed the radar operators that there was no one available since all the officers were at breakfast.

Radar Operations on the day of Pearl Harbor attack

April 27, 2025

1480 words

Page 3

“There’s a massive formation of aircraft advancing towards Oahu from the north, three points east. It’s currently 113 miles out, traveling at 180 miles per hour.” There was a grip of urgency in Elliott’s voice.

McDonald jotted the information down on the nearest piece of paper he could find, along with the time: 7:15. Little did McDonald know that this note would be preserved for the rest of his life and even shared with the public 70 years later. McDonald held the radar operators on the phone and told the report to Lieutenant Kermit Tyler, who was the only officer available. However, even after McDonald urged him twice to take action, Tyler dismissed the report. Tyler knew about six Boeing B-17s expected to arrive from San Francisco this morning. The blip, he thought, must have been those B-17s.

With this in mind, Tyler called the radar operators back. “Don’t worry about it. It’s probably that scheduled flight of B-17s from the mainland.”

Then, without second thoughts, he hung up.

McDonald asked Tyler if they should convey the radar detection to Wheeler Field.

Tyler replied, “Don’t worry about it.”

The lieutenant was mistaken. Those aircraft were not friendly American planes. Like most people, Tyler was unfamiliar with and underestimated the capabilities of radar.

The clock was ticking. Back at Opana Station, Elliott could feel his stomach growl with hunger, prompting him to check the time. It was 7:20. But breakfast didn’t even matter anymore. He was trying to wrap his head around what the mysterious aircraft blip could be. Why were

there so many planes coming all at once? Why was it closing in at such a high speed? Lockard and Elliott kept their eyes on the signal as it advanced closer and closer.

“Why don’t we skip over Tyler and directly contact the Army Pursuit Squadrons?” Elliott suggested uneasily as he made another plot. “They could do something about it.”

“We can’t do that,” Lockard replied, after a moment of pause. “We’re just normal soldiers. We can’t break military protocols—not as enlisted men.”

At 7:40, the radar signal started behaving unstably when it was 22 miles away from the Oahu coastline. By 7:45, the signal was completely lost due to background interference. Elliott stared at Lockard. Lockard stared back at Elliott. Neither person could do anything.

Moments later, the two radar operators closed down the Opana station at 8:00 and were picked up for breakfast. It didn’t take long before the two privates would know what that radar signal meant. People stepped out to see planes emblazoned with red disks thundering overhead and black billows of smoke rising from the direction of Pearl Harbor. Men jumped into the flaming sea water covered with black oil.

The attack on Pearl Harbor had begun.

Japan-U.S. relations grew strained in the 1930s as the Empire of Japan rapidly expanded its influence in Asia and the Pacific. As a response to Japan’s aggressive measures overseas, the U.S. relocated its Pacific Fleet from California to Hawaii in 1940 [1]. In 1939, the U.S. military had started using radar—which stands for “radio detection and ranging”—in its operations. Like

Radar Operations on the day of Pearl Harbor attack

April 27, 2025

1480 words

Page 5

a bat's echolocation, a radar outputs radio waves and senses the echo signals that bounce back from objects. Using this new device, military personnel could identify the position and velocity of targets ahead of time and be prepared.

The radar sets on Hawaii were among the first twenty to be ever used by the U.S. Armed Forces [2]. Radar operators' duty was to track the activity of all surrounding objects and report it to the Fort Shafter information center. Officers at the information center would assess the reports and dispatch pursuit aircraft to intercept potential threats. On December 7, 1941, Radar operators Lockard and Elliott did their job of reporting a massive formation of incoming aircraft. However, when the warning reached Lieutenant Kermit Tyler, it was ignored. The privates below Tyler could not object to Tyler's decision because they had to follow military protocols of obeying their superiors. The neglect of the radar warning resulted in losing the lives of over 2,400 Americans [3].

If Tyler had the chance to better understand the new technology, he could have taken the radar warning seriously. When the U.S. military implements a new tool, they can make sure everyone is updated and on the same page. This will encourage better understanding and communication between military personnel of all ranks. Moreover, Tyler was only on his second day at his new position on December 7. Because the other officers were gone at the same time, Tyler was left to staff the information center by himself. It is better to make sure there is at least one experienced person on duty for each shift.

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Radar Operations on the day of Pearl Harbor attack

April 27, 2025

1480 words

Page 6

What happened at Pearl Harbor can happen anytime and anywhere. This story teaches us that the person in charge must be alert and cautious when making a decision. People should also prioritize having an experienced staff member on duty at all times. During the period leading up to the Pearl Harbor attack, the entire U.S. naval base should have been exceptionally cautious, because everyone was already aware of unwavering Japanese hostility and the possibility of an attack.

On the other hand, keeping up with the latest technology can help us stay at the head of the pack, but human judgement is always the ending factor. Nowadays, even with the widespread use of AI, the person in control has the power of determining the final outcome. So, we must hold ourselves to the highest degree of alertness and caution, especially during critical times. We must be clear minded about current circumstances and know what to expect next. This way, we will be prepared to take the correct steps forward.

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Radar Operations on the day of Pearl Harbor attack

April 27, 2025

1480 words

Page 7

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